

CorporateMind AI

Autonomous AI Corporate Outreach & Multi-Channel Growth OS Multi-Agent AI Operating System for Trainers, Coaches, Consultants, Speakers & Workshop Providers

PRODUCT REQUIREMENTS DOCUMENT — Version **3.0** | 2026 Status: Draft for Build Prepared By: Shyam Sundar G — AI Generalist & Automation Engineer *Confidential | CorporateMind AI | Internal Architecture & Product Specification*

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1. Executive Summary

- **Product Name:** CorporateMind AI
- **One-Line Vision:** An autonomous multi-agent AI platform that converts an independent trainer, coach, consultant, or speaker — from a solo practitioner to a 50-person training agency — into a self-operating B2B growth machine that discovers HR

buyers, generates personalized outreach, runs omnichannel campaigns, follows up, books meetings, drafts proposals, and grows communities with minimal human overhead.

- **Target Users:** NLP coaches, corporate trainers, leadership speakers, wellness coaches, HR trainers, business consultants, workshop providers, motivational speakers; agencies serving these professionals; and white-label resellers (LMS/coaching platforms).
- **High-Level Solution:** A cloud-native, multi-tenant SaaS leveraging a LangGraph multi-agent orchestrator routed through the **Euri AI Gateway** (Claude / GPT / Gemini / DeepSeek / Qwen / Llama / Mistral) on a **free-OSS-first runtime** (FastAPI, PostgreSQL, Redis, Qdrant, optional Ollama) deployable on Vercel + Railway. The platform unifies trainer-profile intelligence, HR discovery, outreach generation, omnichannel campaign distribution, proposal automation, CRM intelligence, and continuous campaign optimization under a single event-driven workflow plane.

Expected Business Impact

Outcome	Baseline (Today)	With CorporateMind AI	Delta
Time to find 100 matching HR contacts	12–20 hours	< 30 minutes	–97%
Cold-outreach reply rate	1.5–3%	12–18%	+500%
Time from poster upload → live campaign	5–7 days	< 60 minutes	–99%
Follow-up cadence completion rate	22% (manual)	96% (automated)	+336%
Corporate meetings booked / trainer / month	1–3	8–14	+400%
Trainer admin hours / week	22	5	–77%
Cost per qualified corporate lead	₹420	₹95	–77%

Revenue Target: ₹2 Cr ARR within 18 months on a 3-tier subscription + usage-metered AI/credits model.

1.1 Product Vision

CorporateMind AI is **not** a CRM, **not** an email tool, **not** a social scheduler. It is the **autonomous AI Business Growth Operating System** that sits underneath all three, replacing the fragmented stack (Apollo + Lemlist + Buffer + Calendly + Notion + a VA + spreadsheets) with a single agentic substrate that observes signals (trainer expertise, HR seniority, campaign performance, channel responses, calendar gaps) and acts on them (segment refinement, outreach drafting, follow-up sequencing, meeting booking, proposal generation) without the trainer prompting it. The trainer stops being the SDR + marketer + ops lead; the system becomes the operator, and the trainer becomes the **approver and the deliverer**.

1.2 Product Name Rationale

CorporateMind fuses the two atomic units of B2B training economics — the **corporate** (the buyer) and the **mind** (the trainer's intellectual asset). The name signals that every trainer's mind is a revenue asset under continuous AI optimization, not a passive document. **AI** as suffix asserts the autonomous, multi-agent posture from first impression.

1.3 Key Differentiators

- **Multi-Agent Orchestrator vs. Single-Model Chatbots:** Specialized agents (RootOrchestrator, TrainerProfile, HRDiscovery, Outreach, Proposal, channel agents for WA/TG/IG/FB/LI/Email, CampaignOptimizer, Analytics, ComplianceGuard, LLMOpsGuardian) coordinate over a shared LangGraph state plane — competitors expose a chat box wrapped around GPT.
- **Content-In, Pipeline-Out:** Trainers upload posters / videos / PDFs / voice intros — AI extracts expertise, niche, tone, pricing, and corporate fit. No 30-field onboarding form.
- **Free-OSS-First Runtime:** Entire MVP runs on free tiers (Vercel + Railway + Qdrant Cloud free + Cloudinary + optional Ollama fallback). Gross margin positive from tenant #1.
- **Provider-Abstracted Inference via Euri Gateway:** Zero vendor lock-in; routing matrix selects the cheapest capable model per task with circuit-breaker fallbacks.
- **Compliance-Native:** Every outbound message passes through ComplianceGuardAgent (opt-in, rate limit, WhatsApp 24h window, dedup, frequency cap, LinkedIn ToS guard) **before** dispatch. No competitor ships this gate as a first-class agent.

- **Tenant-Isolated AI Budgets:** Hard token ceilings, per-feature rate limits, and per-tenant cost telemetry — a control plane competitors lack.
- **HITL by Design:** Sends > 200 recipients, enterprise outreach, and first-week-of-tenant actions are gated. Trainers stay in control without staying in the loop.

2. Problem Analysis

2.1 The Trainer Revenue Leakage Anatomy

Coaches and corporate trainers are skilled at delivery and weak at distribution. ~70% of independent trainers earn under ₹15L/year despite content quality that justifies 3–5×. Revenue is lost at the seam between five disconnected workflows:

Workflow	Today's State	Revenue Leak
HR Discovery	LinkedIn manual scrolling, scraped CSVs	12–20 hrs/week wasted; stale data
Outreach Writing	Templated copy-paste	< 3% reply rate; treated as spam
Follow-up	Manual reminders, forgotten	78% of leads never get touch #2
Channel Mix	Email only; or random IG posts	No omnichannel; weak brand
Proposal Generation	Manual PPT, takes 3–5 hours	Slow turnaround = lost deals

The trainer is the **only integration layer** between these. When the trainer is delivering a workshop, the funnel is paused. CorporateMind AI is the software replacement for the trainer-as-glue.

2.2 Why Current Systems Fail

- **Apollo / Lemlist:** Strong outreach + database but no domain awareness of "training services"; treats every contact as a generic prospect; no follow-up intelligence beyond cadence.
- **HubSpot:** A CRM, not an SDR. Requires a human to drive every action.
- **Zoho/Bigin:** Indian-priced but feature-poor; no AI personalization.
- **Buffer / Hootsuite:** Generic schedulers; no awareness of trainer's expertise or workshop calendar.
- **Generic ChatGPT use:** Stateless, no memory of prior conversations with the same HR, no tool access, no compliance gate.
- **Indian-market trainer tools (Tagmango, Exly):** Built for D2C course selling, not B2B corporate acquisition.

2.3 Why Now

1. **High-reasoning frontier LLMs** (Claude Sonnet 4.6, GPT-5.x, Gemini 3.x) make personalized outreach economically viable at ~₹2–₹8 per fully-customized HR message.
2. **Open-source agentic frameworks** (LangGraph, DSPy) collapse the engineering cost of a 12-agent orchestrator from a 10-engineer year to a 1-engineer quarter.
3. **Post-pandemic L&D budget reset** has created a generation of HR/L&D heads actively shopping for niche specialist trainers — but they do not know who to call.
4. **WhatsApp Business Cloud API maturity** in India unlocks compliant, scalable B2B messaging that did not exist 18 months ago.

3. Market Opportunity

3.1 TAM / SAM / SOM

Layer	Definition	2026 Value	2030 Projection	CAGR
TAM	Global B2B sales-engagement + creator-monetization SaaS	\$48 B	\$112 B	23.6%

SAM	English/Hindi/regional-Asia outreach + creator SaaS for solo experts / agencies	\$7.1 B	\$19 B	27.9%
SOM (5-yr)	India + SEA independent trainers/coaches/consultants and 2–50-person training agencies	\$520 M	\$1.9 B	29.7%

India alone has an estimated ~**3.2 lakh active trainers, coaches, and corporate consultants** with monthly software budgets > ₹2,000 — a **₹760 Cr** addressable monthly opportunity at the entry tier.

3.2 Segment Growth Drivers

Segment	2026 Value	2030 Projection	CAGR
AI Sales Development (AI SDR)	\$2.4 B	\$11.2 B	47%
Outbound Email Automation	\$2.1 B	\$5.8 B	28%
WhatsApp Business Automation	\$0.9 B	\$5.2 B	55%
Creator/Coach SaaS	\$1.6 B	\$6.4 B	41%
AI Proposal Generation	\$0.3 B	\$2.1 B	62%

The **convergence layer** — a single platform that does all five for the trainer/coach segment — has **no category leader**. CorporateMind AI is positioned to define it.

4. User Personas

4.1 Primary Personas

1. Priya (Solo NLP Coach, Bengaluru, 6 yrs experience)

- **Pain:** "I have 14 years of NLP content and I'm still cold-DMing HRs on LinkedIn. I do not know who to email or what to say."
- **Goal:** 2 corporate workshops per month, ₹2L revenue/month, without becoming her own SDR.
- **Channel:** Mobile-first; will not read docs; expects WhatsApp-style UX.

2. Rajiv (Corporate Trainer, Mumbai, 12 yrs, leadership niche)

- **Pain:** Spends Saturdays writing proposals from scratch in PPT. Loses 40% of leads to slow turnaround.
- **Goal:** Same-day proposals; pipeline of 30+ active HR conversations; predictable monthly revenue.
- **Channel:** Desktop + email power user; values customization.

3. Aanya (Founder, 8-trainer agency, Delhi)

- **Pain:** Each trainer markets themselves differently; no central CRM; agency identity is diluted.
- **Goal:** White-label workspace per trainer with shared HR database, central campaigns, brand guardrails.
- **Channel:** Desktop dashboard daily; needs export + admin views.

4. Karthik (Leadership Speaker + Author, multi-city)

- **Pain:** Speaking bookings come through agents who take 30%. Wants direct corporate channel.
- **Goal:** Replace agent dependency; book 4–6 paid speaking gigs/month directly.
- **Channel:** Power user, LinkedIn-heavy; expects polished outputs.

4.2 Secondary Personas

- **The HR Buyer (Meera):** Receives outreach; expects relevance; rebooks if remembered next quarter.
 - **The Agency Operator (Vikram):** Manages multiple trainer accounts; needs RBAC + reporting.
 - **The White-Label Reseller (Edutech SaaS):** Bundles CorporateMind into their LMS for partner trainers.
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5. Product Vision

CorporateMind AI executes a three-horizon vision:

- **Horizon 1 (0–12 months) — The Autonomous Outreach Cockpit.** Replace the trainer's manual SDR + marketer + ops workload. Every HR discovery, message, follow-up, proposal, and campaign is generated, scheduled, executed, and measured by agents. Human role: **approve, edit, deliver.**
- **Horizon 2 (12–30 months) — The Trainer Intelligence Network.** Cross-tenant anonymized signal sharing (which HR titles respond to which message tones; which industries are hiring trainers this quarter) becomes a proprietary data moat. Each trainer benefits from every other trainer on the network.
- **Horizon 3 (30+ months) — The Expert-Services Platform OS.** Open the agent runtime to third-party developers (event organizers, training marketplaces, executive search firms) via a plugin and tool-registration SDK. CorporateMind becomes the agentic substrate that other coaching/training SaaS builds on top of.

6. Business Goals & KPIs

6.1 Business Goals

- **2,000** paying tenants by Month 18.
- 30% MoM growth Months 3–9; 16% MoM Months 10–18.
- **₹2 Cr ARR** by Month 18; **70%** gross margin at scale.
- Net Dollar Retention $\geq$ **115%**.

6.2 Product KPIs

KPI	Target	Why It Matters
Time to first value (TTFV)	< 60 min	Upload → first campaign draft same session
Workflow success rate	$\geq$ 96%	Failed agent run = churn risk
Activation rate	$\geq$ 60%	% of signups that approve their first campaign
90-day retention	$\geq$ 82%	Indian SaaS benchmark is 65–70%
Reply rate on outreach	$\geq$ 12%	4× the industry baseline (~3%)
NPS	$\geq$ 55	Trainers rarely give > 50 unless genuinely impressed

6.3 AI-Specific Metrics

Metric	Target
End-to-end agent run p95 latency	< 15 s (text), < 4 s for inline assistants
Hallucination rate (validated against tools/DB)	< 1.5%
Structured-output schema-validity rate	$\geq$ 99.4%
Self-heal success rate on first retry	$\geq$ 88%
Token COGS per active tenant per month	< ₹180
Semantic cache hit rate	$\geq$ 38%
Drift-detection MTTD	< 24h after model update
HITL override rate	< 8% of agent-proposed actions
ComplianceGuard block rate (false positives)	< 0.5%

7. Scope Definition

7.1 In Scope (Phase 1, Months 0–9)

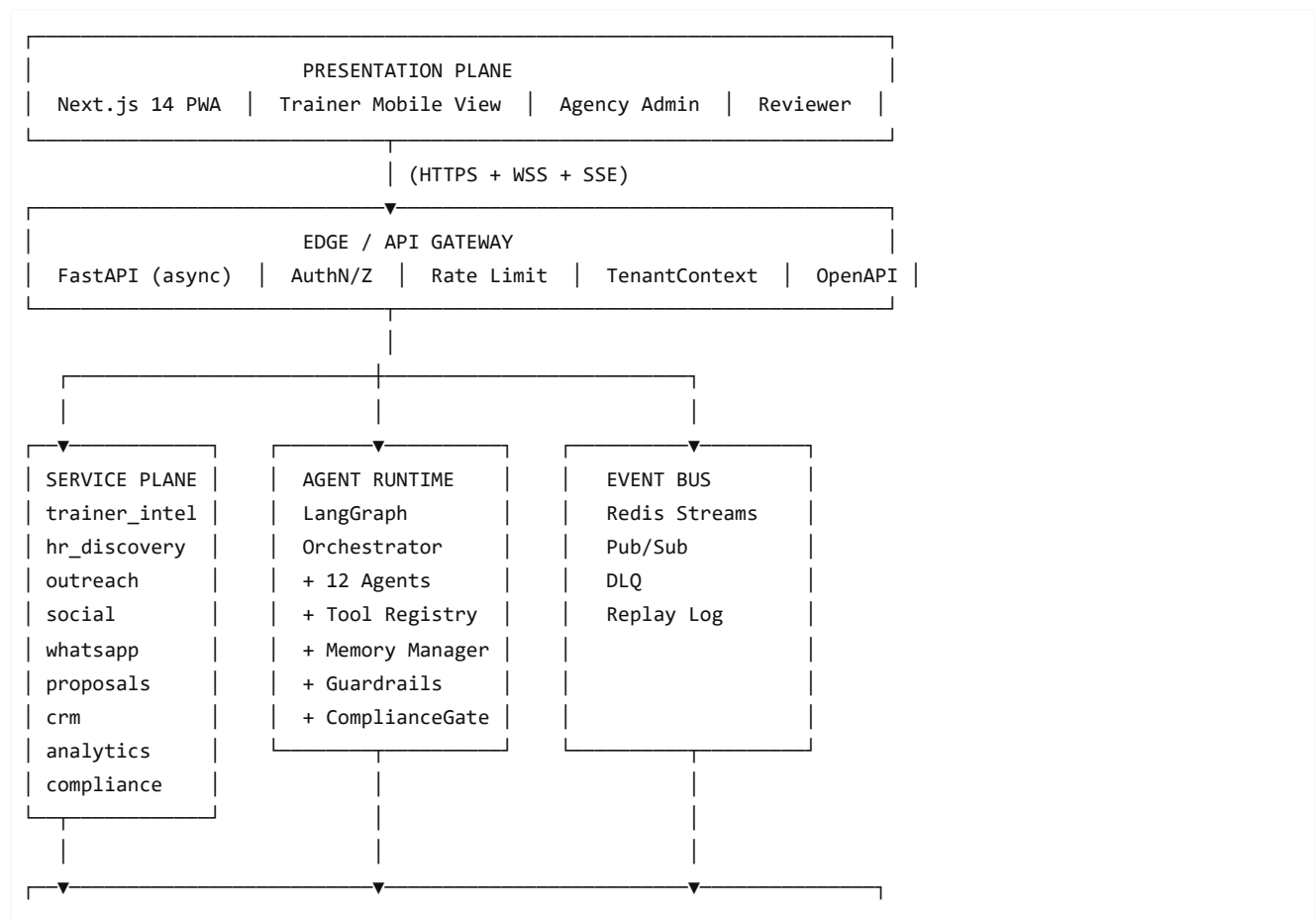
- **12 specialized agents** (full roster in §9)
- **Multi-tenancy:** org → workspace → trainer hierarchy
- **Modules:** trainer intelligence, HR discovery, outreach AI, social automation, WhatsApp engine, proposals, CRM, campaigns, analytics, billing, compliance
- **Channel integrations:** Email (Resend/Postmark/SES), WhatsApp Business Cloud API, Telegram Bot, Instagram Graph (publish + insights), Facebook Graph, LinkedIn (public company data + post publishing only — **no personal DM automation**)
- **Ingestion adapters:** Cloudinary upload + OCR (poster/PDF), audio/video transcription (Whisper self-host or Deepgram free)
- **Self-serve onboarding,** Stripe + Razorpay billing
- **Trainer/Owner dashboard, Agency admin dashboard, Campaign builder, Proposal studio**
- **LLMOps stack:** Langfuse, Promptfoo, DSPy optimization loop

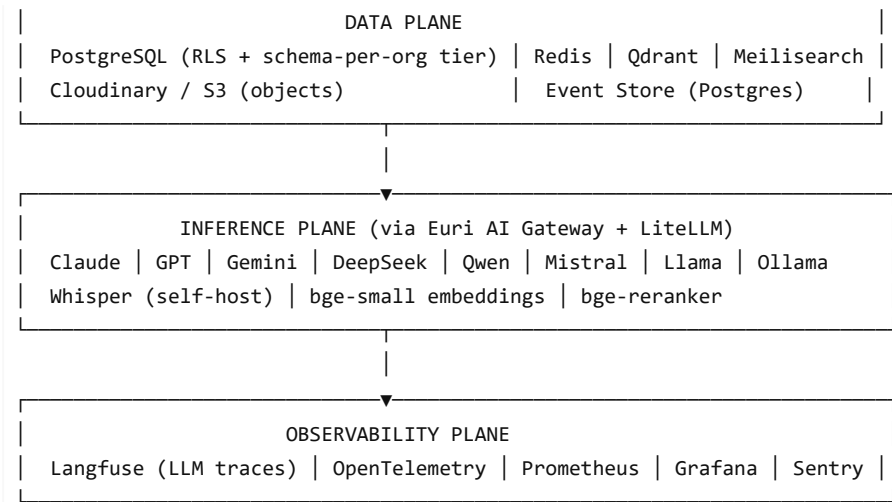
7.2 Out of Scope (Phase 1)

- Inbound voice IVR / outbound voice calls (Phase 2 — reuses pattern from sister product RevenueTable AI)
- Personal LinkedIn DM automation (ToS-risk; permanently out)
- Cold-call dialers
- Course/LMS hosting (we integrate with Tagmango/Exly read-only)
- Payment-collection for trainers (we integrate; we don't disburse)
- Native mobile apps (PWA only)
- On-prem deployment (Enterprise tier only, Phase 3+)

8. Enterprise System Architecture

8.1 High-Level Architecture Plane View





8.2 Multi-Tenancy Model

Hierarchy: Organization (Agency or Solo) → Workspace (brand/trainer) → User (trainer/admin/reviewer) .

Layer	Isolation Strategy	Rationale
Database	Postgres RLS on every base table by <code>tenant_id</code> (default); schema-per-org for tenants > 250 trainers (Enterprise)	Balances cost (RLS cheap) with strong isolation for premium tier
Cache	Redis key prefix <code>t:{org_id}:{workspace_id}...</code> with per-tenant memory budget	Prevents noisy-neighbor cache eviction
Vector	Qdrant collection per org for HR contacts + trainer profiles; shared global collection for prompt cache (PII-stripped)	Tenant data never co-mingled in vector space
Object	Cloudinary path <code>tenants/{org_id}/workspaces/{ws_id}/...</code> with per-tenant signed URLs	Per-tenant lifecycle + retention
Workflow	Celery task routing via <code>tenant_id</code> header; per-tenant queue concurrency cap	One tenant's runaway loop cannot drown shared workers
Inference	LiteLLM virtual key per tenant; per-tenant monthly token ceiling; per-feature rate limit	Hard cost containment; abuse isolation
Observability	Langfuse projects scoped by org; Grafana dashboards filtered by <code>tenant_label</code>	Tenant-specific debuggability without leakage

Entitlements & Feature Flags are resolved at the API gateway via a `TenantContext` middleware. Every downstream service receives a frozen `TenantContext(org_id, workspace_id, plan, ai_budget_remaining, feature_flags, rate_limits)` injected into the request scope. No service queries entitlement state independently — single source of truth at the edge.

8.3 RBAC Inheritance Model

Permissions cascade `Org > Workspace > User` with explicit *deny-overrides-grant* semantics. Built-in roles: `OrgAdmin` , `AgencyManager` , `Trainer` , `Reviewer` , `Analyst` (read-only) . Custom roles supported on Enterprise tier via JSON policy documents (Casbin-style).

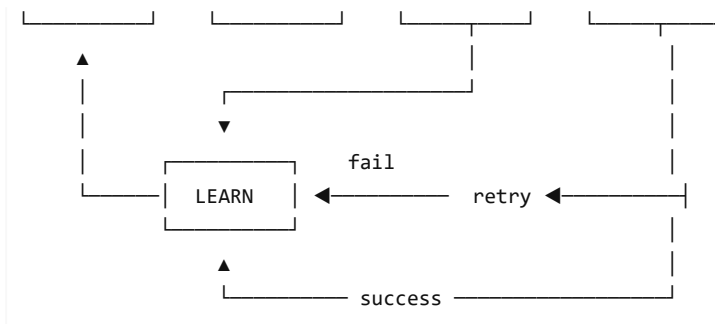
9. Agentic AI Architecture

9.1 Agent Roster

Agent	Primary Role	Tools	Memory Class	Model Tier
RootOrchestrator	Decomposes intent, manages global state, HITL escalator	All downstream agents	Working + Episodic	Mid (Claude Sonnet / GPT-4.1)
TrainerProfileAgent	Extracts niche, topics, tone, pricing, corporate fit from uploads	OCR, Whisper, Vision, Qdrant write	Long-term Semantic	Mid + Vision
HRDiscoveryAgent	Finds matching companies + HR contacts from opt-in sources	Web search, Tavily, public registries, Qdrant	Long-term Semantic	Mid (Claude/GPT for reasoning) + Small (extraction)
OutreachAgent	Drafts per-recipient messages with A/B variants	Trainer profile, HR record, Qdrant retrieval	Working + Episodic	Mid (Claude for copy)
ProposalAgent	Generates pitch decks, agendas, pricing documents	Trainer profile, templates, PDF render	Long-term Semantic	Mid (Claude)
WhatsAppAgent	Template management, 24h window, follow-ups	WA Business Cloud API	Short-term Conversational	Small + rule-based
TelegramAgent	Channel broadcasts, community nurture	Telegram Bot API	Short-term	Small
InstagramAgent	Reel captions, story automation, hashtag intelligence	IG Graph API, image gen	Episodic	Mid (creative)
FacebookAgent	Page publishing, event promotion, messenger	FB Graph API	Episodic	Mid
LinkedInAgent	Public company-page posts, public-data lookups (no DMs)	LI public APIs	Episodic	Mid
CampaignOptimizer	Post-hoc analysis: which segment × channel × tone wins	Analytics DB, Langfuse	Stateless	Small + analytical
AnalyticsAgent	Daily rollups, insight cards, anomaly detection	Postgres analytics, time-series	Stateless	Small
ComplianceGuardAgent	Gate every outbound message: opt-in, rate, dedup, content	Opt-in DB, rate buckets, classifier	Stateless	Small + rules
LLMOpsGuardian	Monitors quality, triggers retries, escalates to HITL	Promptfoo evals, Langfuse, schema validator	Stateless	Small + rule-based

9.2 Reasoning Loop (Per Agent)





- **PERCEIVE:** Hydrate `TenantContext` , load short + semantic memory, normalize input.
- **PLAN:** Generate a structured `PlannerOutput` (Pydantic). LLM forced through `response_format=json_schema` .
- **EXECUTE:** Sandboxed tool executor with per-tool timeout, per-step token budget, idempotency keys.
- **VERIFY:** Schema validation → semantic validators (e.g. "does the HR contact actually exist and is opted-in?") → optional LLM-as-judge.
- **LEARN:** Trace + outcome shipped to Langfuse; periodic batch jobs feed DSPy/Promptfoo eval pipelines.

9.3 Memory Architecture (Summary — full spec in Appendix A)

Memory Class	Backing Store	TTL	Scope	Purpose
Working	In-process dict	Workflow lifetime	Per-run	Plan state, intermediate outputs
Short-term Conversational	Redis conv:{session_id}	30 min sliding	Per-session	Chat assistant turns
Episodic	Postgres agent_runs + agent_events	90 days hot, archive cold	Per-tenant	"Last time we contacted this HR, here's what worked"
Semantic	Qdrant per-tenant collections	Indefinite, with pruning	Per-tenant	Trainer expertise vectors, HR/company vectors, campaign vectors
Procedural (skills)	prompt_templates table	Indefinite	Global + tenant overrides	"How to draft a wellness-coach outreach to a 5000+ employee tech firm"

9.4 Guardrails

- **Input guardrails:** PII redaction (Presidio rules + regex), prompt-injection classifier (LLM-Guard), tenant-scope enforcement on tool calls.
- **Output guardrails:** JSON-schema validation, fact-grounding (output must cite tool/DB IDs when claiming a fact), regex denylists, profanity + competitor-name moderation.
- **Action guardrails:** Any agent action affecting > 200 recipients auto-routes to HITL. Any campaign cost > tenant threshold triggers approval webhook. First-week-of-tenant: everything HITL by default (**training-wheels mode**).

10. Autonomous Workflow Design

10.1 LangGraph State Machine

Every workflow is a directed graph with:

- Typed `WorkflowState` (Pydantic) as the single source of truth between nodes.
- Nodes are pure functions `(state) -> partial_state_update` .
- Edges are conditional functions returning the next node name or `END` .
- Every transition is checkpointed to Postgres (`workflow_checkpoints`) keyed by `(workflow_id, node_id, version)` .

10.2 Workflow Persistence & Resumption

A workflow can be paused at any node, persisted, and resumed hours later — critical for HITL gates ("trainer must approve this campaign before send"):

```
Event → Wakeup Handler → Load Checkpoint → Validate State Schema Version
      → Resume LangGraph from last node → Continue
```

If the state schema evolved (rare), a migration function registered per (from_version, to_version) pair runs on hydration. Workflows older than 30 days depending on a removed node are quarantined and surfaced for manual closure.

10.3 Retry, Backoff & Circuit Breakers

Failure Class	Strategy
Transient (network, 5xx from Euri)	Exponential backoff: 1s, 3s, 9s; max 3 attempts
Rate-limit (429)	Token-bucket-aware retry with jitter; fall back to next model in routing matrix
Schema-invalid output	Self-repair: re-prompt with parsing error appended; max 2 retries; then escalate
Tool execution failure	Idempotency-key replay; if same failure twice, mark node failed and route to recovery
LLM hallucination (grounding fails)	Re-run with stricter system prompt + temperature 0; if still fails, HITL
Tenant budget exceeded	Fail fast, raise BudgetExceeded, notify org admin, do not retry
Channel provider failure (WA template rejected, IG API down)	Mark recipient failed_transient; reschedule with backoff; circuit-break after threshold

Circuit breakers wrap every external dependency (Euri, WhatsApp, Resend, Telegram, IG, FB, LinkedIn, Cloudinary). Half-open after 60s; metric circuit_state{dep} exposed to Prometheus.

10.4 Dead-Letter Queue & Replay

Failed workflows land in dlq_workflows with full state, error fingerprint, trace ID. Daily reaper groups failures by fingerprint and surfaces top-10 patterns to LLMOpsGuardian. Replays triggered manually via admin API or programmatically once a fix ships.

11. Feature Breakdown (Functional Requirements)

Feature 1 — Trainer Intelligence (AI Profile Extraction)

User Story: As a trainer, I upload my posters/videos/PDFs/voice intro and the platform builds a complete profile of my expertise without me filling a form.

Acceptance Criteria:

- Accepts: PDF, JPG/PNG, MP4, MOV, MP3, WAV, public URL
- Extracts: niche, workshop topics (multi-tag), industries served, experience years, communication tone, pricing hints, CTA, target seniority, language
- Produces: trainer profile vector (Qdrant) for semantic matching
- User can review + edit every extracted field before lock-in

Edge Cases & Constraints:

- **Vision pipeline:** Posters in Hindi/regional scripts handled via multilingual OCR.
- **Audio in Hinglish/code-switch:** handled natively, no translation hop.

- **Conflicting signals across uploads:** the latest upload wins; user prompted to resolve.
- **Sensitive content:** if upload contains PII other than the trainer's own, surfaced for redaction approval.

Feature 2 — HR Discovery Engine

User Story: As a trainer, I want the system to find HR managers, L&D heads, and wellness leads at companies that match my expertise — without me building a list.

Acceptance Criteria:

- Discovers HR contacts from **opt-in / public sources only:** public company directories, company career pages, webinar registrations the trainer brings, public LinkedIn company pages (company-level data only, no scraping private profiles)
- Match score (0–1) per (trainer × company) using semantic similarity + rules
- Default ≥ 50 matched contacts within 30 minutes of profile lock
- Industry, size, region, role-level filters
- Dedup across the org

Edge Cases & Constraints:

- **Sources allowlist:** ComplianceGuard maintains an explicit allowed-sources registry; any scraper attempting an unlisted domain is rejected.
- **Opt-in proof:** Every contact stores `source` , `source_type` , `opted_in_at` , `opt_in_evidence` (URL, screenshot ref, consent timestamp).
- **Stale data:** Contacts older than 6 months are flagged and de-prioritized in segments.
- **PII:** Phone is hashed in analytics tables (HMAC with per-org salt); email stored as-is for delivery, redacted in logs.

Feature 3 — AI Outreach Generator

User Story: As a trainer, I want every email, WhatsApp, and LinkedIn-post outreach to feel personally written for that specific HR person.

Acceptance Criteria:

- Per-recipient personalization referencing: their company industry, recent public news (opt-in source), trainer's matching expertise
- Channel-aware copy: Email (subject + preheader + body), WhatsApp (≤ 350 chars, emoji-aware), LinkedIn post (≤ 1200 chars), Telegram (multi-paragraph allowed)
- 2 variants per recipient for A/B
- Tone variants: formal / consultative / friendly
- Optimal send-time prediction per segment per channel
- Suppression list automatically enforced

Edge Cases & Constraints:

- **WhatsApp template compliance:** Marketing templates pre-cleared with Meta; template versioning + approval status tracked; non-approved templates blocked at send time.
- **Rate limits:** WA Business API tier-aware throttler; never exceed tier.
- **Cooldown:** No HR contact receives > 2 marketing messages in 7 days across all channels (cross-channel cap).
- **Hallucination check:** Any claim about the trainer's credentials must resolve to the trainer's locked profile — otherwise blocked.
- **HITL gate:** Sends > 200 recipients require trainer approval; sends > 1,000 require OrgAdmin.

Feature 4 — Omnichannel Social Automation

User Story: As a trainer, I want my content distributed across my channels with the right copy for each platform — without me logging in to five apps.

Channels & Capabilities:

Channel	Capabilities
---------	--------------

Email	Personalized campaigns, sequence cadence, reply-detection
WhatsApp	Group outreach, broadcast lists, template send, follow-up flows
Telegram	Channel broadcast, community nurture, webinar drip
Instagram	Reel captions, story automation, carousel generation, hashtag intelligence, post scheduling
Facebook	Page publishing, event promotion, Messenger workflows
LinkedIn	Public company-page posts only (no personal DM automation)

Feature 5 — WhatsApp Automation Engine

Capabilities: Official WA Business Cloud API integration, AI outreach, HR follow-ups, webinar campaigns, group invite distribution, smart segmentation, campaign optimization.

IMPORTANT COMPLIANCE:

- ☒ Official WA Business Cloud API only
- ☒ Opt-in tracking per (tenant, contact)
- ☒ Rate limiting per WA tier
- ☒ Anti-spam (frequency cap, cooldown)
- ☒ Unsubscribe management (global per tenant)
- ☒ 24-hour customer-care window enforcement (templates required outside)

Workflow:

Workshop Upload → AI Generates Campaign → AI Finds Matching HRs
→ Campaign Approval → Rate-Limited Distribution → AI Follow-ups

Feature 6 — Telegram Automation

Bot integration, auto-posting, webinar broadcasts, AI engagement campaigns, community nurturing, inline keyboards for replies.

Feature 7 — Instagram & Facebook Automation

Instagram: AI reel captions, story automation, hashtag intelligence, carousel generation, post scheduling. Facebook: Page publishing, campaign automation, event promotion, Messenger flows.

Feature 8 — AI Proposal Generator

User Story: As a trainer, after a positive HR reply, I want a customized pitch deck + agenda + pricing in under 2 minutes.

Acceptance Criteria:

- Input: HR conversation context + trainer profile + workshop selection
- Output: PDF (via WeasyPrint), Google Doc, Notion (optional)
- Sections: cover, problem framing, solution, agenda, deliverables, pricing, case studies, next steps
- Trainer can edit any section in-app before send

Feature 9 — CRM & Lead Intelligence

Tracks: leads, meetings, replies, open rates, click rates, conversion, industry analytics, AI campaign performance. Pipeline board (Kanban): discovered → contacted → engaged → replied → meeting_scheduled → proposal_sent → booked | lost .

Feature 10 — Multi-Agent AI System

(Spec covered in §9, §10, §27, Appendix B.)

Feature 11 — LLMOps & AI Observability

12. User Journey Flows

12.1 Trainer Onboarding (Day 0 → Day 1)

Signup → Email/Phone OTP → Org creation → Workspace wizard (name, niche, language)
→ Upload assets (poster/video/PDF/voice) → AI extracts profile (<5 min)
→ Trainer reviews + edits profile → Connect channels (WA, Email, IG)
→ AI proposes first 50 HR matches (<30 min) → Trainer approves segment
→ First campaign drafts shown → Trainer approves → First sends within Hour 1

12.2 Autonomous Campaign Loop

Cron (08:00 daily) → AnalyticsAgent computes momentum → CampaignOptimizer proposes
3 next-action campaigns (re-engage Lapsed, nurture Engaged, expand to new industry)
→ ComplianceGuard validates → Trainer inbox card ("Approve / Edit / Reject")
→ On approve → Send pipeline (batched, throttled, cross-channel cap respected,
attribution links injected) → Outcome events streamed back → 72h post-send
attribution report → Outcome ingested into RLHF feedback store

12.3 HR Reply → Proposal → Meeting Flow

Inbound reply (Email / WA / Telegram) → Reply classifier (positive / neutral / negative
/ unsubscribe) → If positive → Conversation thread surfaced to trainer → Trainer triggers
"Generate proposal" → ProposalAgent drafts (<2 min) → Trainer edits + sends →
Calendly link auto-embedded → Booking webhook → meeting created → reminder sent →
post-meeting summary requested

12.4 Compliance Gate Flow

Any outbound message → ComplianceGuardAgent:
- check opt-in for (contact, channel)
- check unsubscribe list
- check frequency cap (≤ 2 msgs / 7 days / cross-channel)
- check WA 24h window if WA template
- check content policy classifier
- check tenant budget
→ Pass: queue for send | Block: log reason + notify trainer | HITL: route to approval

13. API Specifications & Governance

13.1 API Surface

- **Public REST API** (/api/v1/*): tenant-scoped, OAuth2 + JWT, OpenAPI 3.1 documented.
- **Internal gRPC** (Phase 2): agent-to-service for hot paths.
- **WebSocket** (/ws/v1/agent-stream): live LangGraph events to dashboard.
- **SSE** (/sse/v1/insights): one-way push for insight cards.
- **Webhooks (outbound)**: tenant-registered endpoints receive `lead.*`, `campaign.*`, `proposal.*`, `meeting.*` events.
- **Webhooks (inbound)**: WhatsApp, Telegram, IG, FB, Calendly, Stripe/Razorpay — all HMAC-verified.

13.2 Versioning & Lifecycle

URI versioning (`/api/v1` , `/api/v2`). **Never break a v.** Additive changes only within a major. Deprecation: 6-month sunset window with `Deprecation` + `Sunset` headers. Breaking changes ship behind `?api_preview=v2` opt-in for 90 days minimum.

13.3 Contracts & Schema Evolution

- All request/response bodies are Pydantic models → OpenAPI exported on every CI build.
- Schema-diff job in CI compares against previous spec; non-additive change requires PR label `api:breaking` and architect approval.
- SDK generation (TypeScript + Python) on every merge to `main` .

13.4 Idempotency & Pagination

- Every mutating endpoint accepts `Idempotency-Key` header; server stores result 24h.
- Pagination: cursor-based only (`cursor` , `limit` , default 50, max 200). Offset pagination forbidden by lint rule.

13.5 Webhook Standards

HMAC-SHA256 signed with `X-CM-Signature` header. Retries: 1m, 5m, 15m, 1h, 6h, 24h (jittered). Permanent failure after 6 attempts → DLQ + admin notification. At-least-once delivery; consumers must be idempotent on `event_id` .

13.6 Rate Limits (Default Tier)

Endpoint Class	Starter	Growth	Enterprise
Read	60 rpm	300 rpm	Custom
Mutating	30 rpm	150 rpm	Custom
Agent-trigger	10/hour	100/hour	Custom
Outreach sends/day	500	5,000	Custom

Enforced via Redis token bucket keyed on `(tenant_id, endpoint_class)` .

14. Database Design

14.1 Core Schema (PostgreSQL 16)

Table	Purpose	Key Columns
orgs	Top-level tenant	id, name, plan, region
workspaces	Brand/trainer within org	id, org_id, brand_name, niche, voice_profile_id
users	Operators	id, org_id, email, role, mfa_enabled
trainers	Trainer profile	id, workspace_id, name, niche_tags, tone, pricing_range, experience_yrs
uploads	Posters/videos/PDFs	id, trainer_id, kind, status, cloudinary_url, ocr_text, transcript
companies	Discovered corporates	id, name, industry, size_band, region, source
hr_contacts	HR people	id, org_id, company_id, name, title, email_hash, phone_hash, opt_in_status, source
lead_lists	Saved segments	id, workspace_id, name, query_json

campaigns	Campaign lifecycle	id, workspace_id, channel(s), status, audience_query, schedule_at, approval_state
campaign_recipients	Per-recipient state	id, campaign_id, hr_contact_id, status, sent_at, replied_at
outreach_messages	All sent messages	id, channel, body, variant, sent_at, delivery/open/reply status
conversations	Threaded replies	id, hr_contact_id, channel, last_msg_at
proposals	Pitch decks	id, conversation_id, status, version, pdf_url
meetings	Calendar events	id, conversation_id, provider, scheduled_at, summary
whatsapp_messages, telegram_posts, instagram_posts, facebook_posts, linkedin_posts	Per-channel logs	...
agent_runs	Workflow executions	id, tenant_id, agent_name, status, tokens, cost_cents
agent_events	Per-node events	id, run_id, node, payload_json, trace_id
workflow_checkpoints	Resumable state	workflow_id, node, state_json, schema_version
events	Domain event log	id, tenant_id, type, payload_json, occurred_at, source, version
dlq_workflows	Failed workflows	id, run_id, error_fingerprint, state_json
prompt_templates	Versioned prompts	id, name, version, body, eval_score, active_in_envs
opt_ins, unsubscribes	Compliance	id, tenant_id, contact_id, channel, evidence
audit_events	Append-only audit	id, tenant_id, actor, action, target, occurred_at
billing_plans, subscriptions, usage_counters, invoices	Monetization	...

14.2 Indexing & Partitioning

- outreach_messages : native partitioning by RANGE (sent_at) monthly; auto-detach > 18 months to cold S3 Parquet.
- agent_events : hot-cold split — last 14 days in Postgres, older shipped to R2/S3 Parquet, queryable via DuckDB.
- events : append-only; weekly compaction.
- Composite (tenant_id, created_at DESC) on every event-style table.
- GIN on JSONB metadata.

14.3 Multi-Tenant Isolation

- Small/mid tenants: RLS policies USING (tenant_id = current_setting('app.tenant_id')::uuid) . Tenant ID set per connection via SET LOCAL app.tenant_id =
- Enterprise tier: dedicated schema-per-org with logical replication for analytics.

15. Vector Database Strategy

15.1 Qdrant Collections

Collection	Vectors	Payload	Tenancy
------------	---------	---------	---------

trainer_profiles_{org_id}	768-d bge-small	trainer_id, niche_tags	Per-org
companies_{org_id}	768-d	company_id, industry, size	Per-org
hr_contacts_{org_id}	768-d	contact_id, title, seniority	Per-org
campaign_outcomes_{org_id}	768-d	campaign_id, segment, channel, outcome	Per-org
prompt_cache_global	1024-d	prompt_hash, response_ref, hit_count, ttl	Cross-tenant (semantic cache only — no PII)
proposal_templates_{org_id}	768-d	template_id, vertical	Per-org

15.2 Retrieval Pipeline

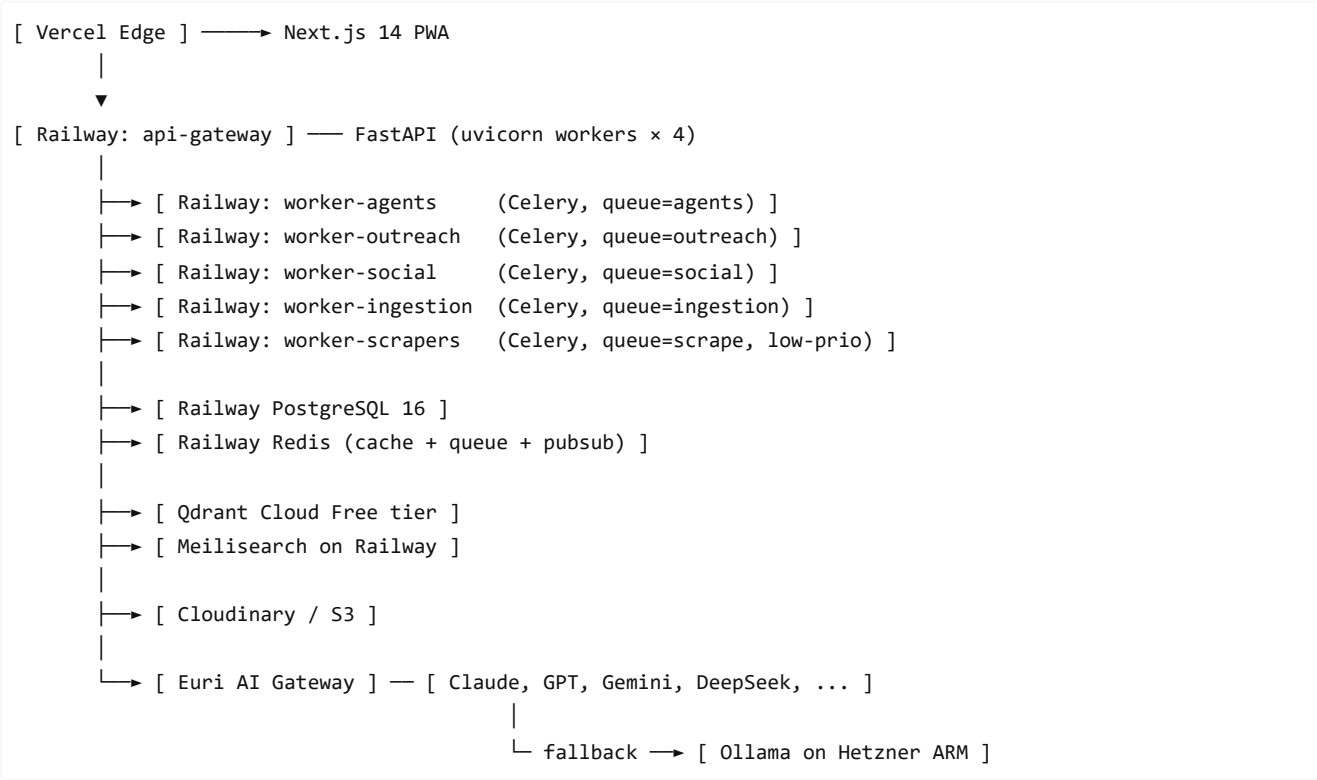
Query → Embed (bge-small via local sentence-transformers) → Hybrid Search (Qdrant ANN top-50 + Meilisearch BM25 top-50) → RRF Fusion → Cross-encoder Rerank (bge-reranker-base) → Top-k (default 8) → Inject into prompt with provenance IDs

15.3 Memory Pruning

- Trainer profile vectors: re-embed on every profile lock + quarterly refresh.
- HR contact vectors: 12-month rolling window; stale flag at 6 months.
- Campaign outcome vectors: retain indefinitely; this is the RLHF moat.
- Prompt cache: LFU eviction with 30-day TTL hard ceiling.

16. Infrastructure Architecture

16.1 Topology (MVP — Stage 1)



16.2 Real-Time

- **Redis Pub/Sub** for fan-out (agent streaming tokens to dashboard).
- **WebSocket** server colocated with FastAPI (Redis subscriber bridge).
- **SSE** for one-way insight cards.
- **No Kafka** in Stage 1. Trigger for adoption: > 10k events/sec sustained.

16.3 Storage Tiers

Tier	Store	Use
Hot OLTP	Postgres	Live transactional state
Hot Cache	Redis	Session, semantic cache, rate-limit buckets
Vector	Qdrant	Semantic memory
Object	Cloudinary / S3	Uploads, generated PDFs, exports
Cold Analytics	R2/S3 Parquet + DuckDB	Long-tail event log, historical messages

17. Deployment Strategy

17.1 Environments

dev (local Docker Compose) → preview (per-PR Vercel + Railway) → staging (mirrors prod, daily snapshot restore) → prod .

17.2 CI/CD (GitHub Actions)

PR opened → lint (ruff, eslint, mypy, tsc) → unit tests → OpenAPI diff check
→ alembic upgrade check → build Docker (multi-stage, distroless final)
→ deploy preview → e2e smoke (Playwright)

PR merged → deploy staging → run Promptfoo eval suite (gate)
→ canary 10% prod → metric/error budget check (15 min)
→ full rollout → tag release

17.3 Database Migrations

Alembic, one migration per PR, **expand-then-contract**:

1. Additive migration (add column, default null)
2. Backfill job (idempotent, batched)
3. App version that reads + writes new column
4. Cleanup migration (NOT NULL, drop old)

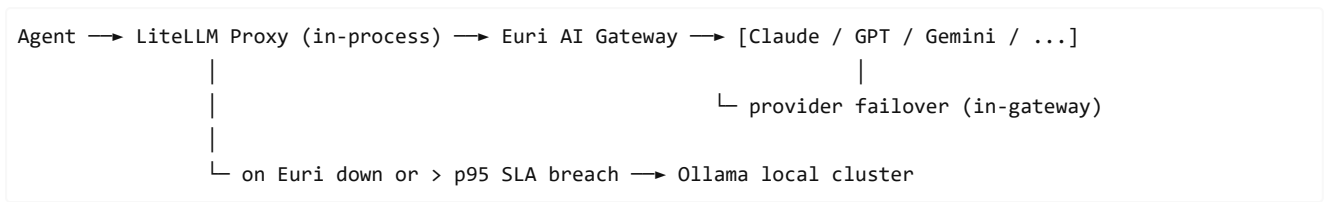
Never deploy step 4 in same release as step 1.

17.4 Rollback

- Vercel: instant via dashboard.
- Railway: previous image tag pinned, `railway rollback` .
- DB: forward-only philosophy; rollback via compensating migration.

18. LLMOps Architecture

18.1 Inference Gateway Topology



18.2 Model Routing Matrix

Task	Primary	Secondary	Local Fallback
Outreach copy (personalized)	Claude Sonnet	GPT-4.1	Llama 3.3 70B
Long-context planning	Claude Sonnet	GPT-4.1	Qwen 2.5 32B
Structured extraction (JSON)	DeepSeek V3	Gemini Flash	Qwen 2.5 14B
Social captions / creative	Gemini Flash	Claude Haiku	Llama 3.3
Lead classification / ranking	DeepSeek	Qwen 14B	Phi-3.5
Proposal generation	Claude Sonnet	GPT-4.1	Llama 3.3 70B
Reply intent classification	GPT-4o-mini	Gemini Flash	Phi-3.5
Embedding	bge-small (self-hosted)	OpenAI text-embedding-3-small	—
Reranking	bge-reranker-base (self-hosted)	Cohere Rerank	—
Moderation	Llama Guard 3 (self-hosted)	OpenAI Moderation	—

Routing decision per request based on `task_class` , `tenant_plan` , `cost_budget_remaining` , `latency_target` .

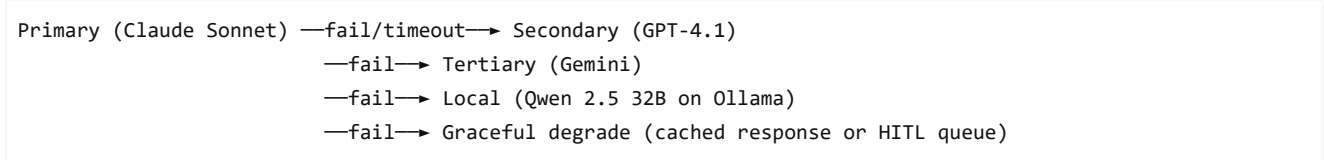
18.3 Semantic Cache

Input prompt → embed → check `prompt_cache_global` for cosine ≥ 0.96 → if hit + same `task_class` + same `model_tier` → return cached response. Cache only deterministic-by-input tasks (extraction, classification). **Never cache personalized outreach copy** (would defeat personalization). Per-tenant private cache for prompts containing tenant data. PII filtered by Presidio before hash.

18.4 Prompt Optimization

- All prompts versioned in `prompt_templates` ; promotion to `active` requires Promptfoo eval pass.
- Weekly cron: DSPy `BootstrapFewShot` over last 7 days of high-rated outcomes to propose new templates.
- Candidate vs. champion A/B at 10% traffic for 72h; promote on score lift > 5% and no guard regression.

18.5 Fallback Chains



Every fallback decrement logged with reason; SRE alert on > 3% fallback-to-local rate over 5-min window.

18.6 Eval Pipelines

- **Pre-deploy gate (Promptfoo):** 200+ test cases per agent, run on every PR touching prompts; merge blocked on regression > 2%.
- **Continuous eval (daily):** sampled production traces re-scored by LLM judge for groundedness, format compliance, tone match.
- **Adversarial eval (weekly):** prompt-injection battery, jailbreak attempts; feed guardrail rule updates.

19. Monitoring & Observability

19.1 Stack

- **Langfuse** — LLM trace store; every agent run = one trace with nested spans per node/tool/model call.
- **OpenTelemetry** — application + workflow spans; `traceparent` propagated HTTP → Celery → DB.
- **Prometheus** — system + custom business metrics on FastAPI `/metrics` + Celery exporter.
- **Grafana** — dashboards (see §19.3).
- **Sentry** — error capture, release tracking, performance.

19.2 Correlation Strategy

Every inbound HTTP request gets a `request_id` (ULID). Propagated: HTTP → Celery task headers → LangGraph `RunnableConfig.metadata` → Langfuse trace `metadata.request_id`. One ID joins HTTP logs ↔ DB query logs ↔ Celery logs ↔ LLM trace.

19.3 Dashboards

Dashboard	Panels
Tenant Health	Per-tenant request rate, p95 latency, error %, token spend, agent success %
Agent Runtime	Run count by agent, success rate, p50/p95/p99 latency, retry rate, HITL rate
LLMOps	Token usage by model, cost projection, fallback rate, cache hit rate, eval score trend
Workflow	DLQ size, replay rate, checkpoint count, longest-paused workflows
Channels	Per-channel send rate, delivery %, reply %, compliance-block rate
Business	New tenants, MRR, churn, avg reply rate, avg meetings/tenant/month

19.4 SLOs & Alerts

SLO	Target	Burn Alert
API availability	99.9% monthly	2% budget burn in 1h → Sentry alert
Agent run success	96% rolling 24h	< 94% for 30 min → page on-call
Outreach send error rate	< 1%	> 2% for 15 min → page
Token cost / tenant / day	< ₹20 avg	> ₹40 for any tenant → notify ops
ComplianceGuard false-positive rate	< 0.5%	> 1% for 1h → notify

19.5 Workflow Replay Debugging

Any failed run is fetchable via admin UI with full state diff per node and re-executable against staging with the exact same `WorkflowState` + sandbox tool registry — same input, deterministic replay.

20. Security Architecture

- **AuthN:** Email/Phone OTP for trainers; OAuth2 + JWT (RS256, 15-min access + 30-day refresh); MFA enforced for OrgAdmin+.
- **AuthZ:** Casbin policy engine; deny-overrides; policies version-controlled.
- **Secrets:** Doppler / Infisical (prod); Railway Secrets (preview); rotated 90-day; CI fails on `gitleaks` finding.
- **Transport:** TLS 1.3 everywhere; HSTS preload; mTLS between internal services (Stage 2+).
- **At-Rest:** Postgres TDE; Cloudinary/S3 SSE; secrets sealed.

- **PII Masking:** Presidio analyzer on every prompt pre-model; redacted tokens substituted; reversed only in tenant-scoped post-processing.
- **Prompt Injection:** LLM-Guard input filter; output validators with allowlist for tool invocation; no agent can call an unregistered tool.
- **Output Validation:** Structured outputs schema-validated; free-text outputs pass moderation + grounding (must cite tool IDs when claiming facts).
- **Rate Limits:** Per-tenant, per-endpoint, per-IP layered; abusive IP auto-blocked at Cloudflare WAF.
- **Audit:** Append-only `audit_events` for every privileged action; 7-year retention for Enterprise.
- **Compliance:** GDPR + India DPDP Act — right to erasure as cascading soft-delete + 30-day hard purge; per-tenant data residency tag; per-tenant data export endpoint.

21. AI Safety & Governance

- **Bias mitigation:** Audience selection cannot include/exclude on demographic attributes not explicitly consented. Forbidden filters (gender, religion, caste): hard denylist; lint at policy-creation.
- **Content moderation:** Llama Guard 3 on every outbound customer-facing message; block + alert on flagged categories.
- **Hallucination floor:** Any claim about a trainer's credentials, pricing, or workshop content must resolve to a locked profile row; unresolved claims trigger re-prompt or HITL.
- **HITL gates:**
 - Campaign sends to > 200 recipients
 - Any agent action with monetary impact > ₹2,000
 - First-week-of-tenant: everything HITL by default (**training-wheels mode**)
 - Any LinkedIn/Email/WA send to a "do-not-contact" flagged contact (defense-in-depth)
- **Explainability:** Every AI-proposed action carries a `rationale` field (≤ 280 chars) shown in UI.
- **Right to opt-out of AI training:** Tenants can opt out of contributing data to the cross-tenant model loop while still receiving model improvements.

22. Cost Estimation & Token Economics

22.1 Per-Tenant AI Cost Model (Growth Plan)

Workload	Runs/Month	Avg Tokens (in+out)	Avg Cost @ Mid-tier (₹)	Subtotal
Trainer profile extraction (one-time + edits)	8	18,000	2.40	₹19
HR discovery runs	12	28,000	3.80	₹46
Outreach generation	2,500	1,800	0.30	₹750
Reply classification	800	600	0.05	₹40
Proposal generation	35	22,000	3.20	₹112
Social caption / creative	120	1,400	0.18	₹22
Analytics + optimizer (daily)	30	12,000	1.60	₹48
Guardrails + evals overhead	—	—	—	₹160
Subtotal pre-cache				₹1,197
Semantic cache savings (~38%)				–₹455
Net AI COGS / tenant / mo				₹742

Target Growth-tier ASP ₹4,999/mo → **gross margin ~85% on AI alone**, before infra amortization. (Higher than RevenueTable because text-only — no voice COGS.)

22.2 Cloud Infrastructure

Component	100 Tenants	1,000 Tenants	10,000 Tenants
Vercel (frontend)	₹0	₹1,800	₹14,000
Railway (compute + Postgres + Redis)	₹4,500	₹38,000	₹2,80,000
Qdrant	₹0	₹6,500	₹52,000
Cloudinary / S3	₹400	₹3,500	₹32,000
Euri AI inference (net of cache)	₹74,200	₹7,42,000	₹74,20,000
Observability (Langfuse self-host + Sentry)	₹1,000	₹8,000	₹65,000
Total / month	₹80,100	₹7,99,800	₹78,63,000
Per-tenant / month	₹801	₹800	₹786

Per-tenant cost is nearly flat — infra scales linearly with inference; everything else amortizes.

22.3 AI Budget Enforcement

- Per-tenant monthly token ceiling in `TenantContext.ai_budget` .
- Pre-call estimator on every LLM call; if `spent + estimate > budget` , queued (best-effort) or rejected (hard ceiling).
- 70% / 85% / 95% threshold alerts to OrgAdmin via email + dashboard banner.
- Premium models (Claude/GPT-4-class) reserved for: personalized outreach, proposals, strategic copy. Cheap models (DeepSeek, Qwen, Gemini Flash, Haiku) for: classification, extraction, ranking.

23. Scaling Strategy & Infrastructure Evolution

23.1 Stage 1 — Solo-Dev MVP (0–500 tenants)

Modular monolith, 5 Celery worker pools, single Postgres + Redis, Qdrant free tier, optional Ollama on Hetzner ARM (\$14/mo) for fallback. **Bottleneck:** Postgres connections + Celery concurrency.

23.2 Stage 2 — Multi-Worker, Service-Split (500–5,000 tenants)

Extract `social/` and `whatsapp/` modules as separate services (I/O-heavy). Postgres read-replica for analytics; PgBouncer for connection pooling. Redis clustered. Qdrant paid tier with replication. Cloudflare edge cache. Full OTel + Grafana stack. **Trigger:** p95 > 800ms at gateway OR Celery queue depth > 200 sustained.

23.3 Stage 3 — Distributed, Multi-Region (5,000+ tenants)

Kubernetes (Hetzner/EKS managed). Postgres sharded by `org_id` (Citus) or per-region cluster. Kafka/Redpanda for cross-service event bus; Redis Pub/Sub retained intra-service. Multi-region inference. Dedicated GPU pool for self-hosted Llama 70B + reranker. SOC2 Type II audit complete. **Trigger:** ARR > ₹4 Cr OR first Enterprise dedicated-VPC ask.

23.4 Migration Tradeoffs (Honest)

- **Premature K8s** is the #1 failure mode of agentic startups — defer until ≥ 3 engineers + clear scaling pain.
- **Premature Kafka** = 6 weeks of ops work for benefit only > 10k EPS.
- **Premature multi-region** introduces consistency headaches; defer until a paying tenant requires it.

24. Business Model

24.1 Pricing Tiers

Tier	Starter	Growth	Enterprise	White-Label
Price (INR/mo)	₹999	₹4,999	₹19,999+ (custom)	Custom
AI runs/month	1,000	15,000	Unlimited (fair-use)	Per agreement
Outreach sends/month	500	5,000	Unlimited	Per agreement
Agents enabled	Trainer, HRDiscovery, Outreach, Email + WA	+ IG, FB, Telegram, Proposal	All + custom	All + branded
Channels	Email, WhatsApp	+ IG, FB, Telegram, LinkedIn-post	+ Custom integrations	All
Workspaces	1	5	Unlimited	Unlimited
Support	Email	Email + chat	Dedicated CSM	Dedicated team
Data residency	Shared India	Shared (India / US)	Dedicated cluster	On-prem option

24.2 Add-Ons

- Extra outreach sends: ₹0.40 / message
- Extra AI runs: ₹40 / 1,000 runs
- Custom integration setup: ₹50,000 one-time
- White-label setup: ₹1,50,000 one-time + ₹25,000/mo per reseller

24.3 CAC / LTV Assumptions

- Blended CAC: ₹3,200 (organic content + community + paid)
- Average ARPU: ₹3,800/mo
- Gross margin: 75%
- Logo churn: 2.0%/mo target → 14-month payback
- **LTV/CAC target: ≥ 4x** within 18 months

25. ROI Analysis (Trainer's POV)

Lever	Baseline	With CorporateMind AI	Annualized Impact (solo trainer)
HR research hours saved	16 hrs/week	1 hr/week	+780 hrs/yr
Outreach reply rate (2% → 14%)	60 leads × 2% = 1.2 mtgs	60 leads × 14% = 8.4 mtgs	+7.2 meetings/cycle
Workshops booked / month	1	4	+36 workshops/yr @ ₹40k = + ₹14.4 L
Proposal turnaround (4h → 5m)	12 proposals/yr lost to slow turnaround	All same-day	+6 closes/yr @ ₹40k = + ₹2.4 L
Marketing tool stack replaced	₹8k/mo (Apollo + Buffer + Calendly + ...)	₹5k/mo CorporateMind	– ₹36 k/yr cost
Total annual impact			~₹17 L net upside

CorporateMind cost (Growth, 12 mo)			~₹60 k
Net			~₹16.4 L → 27× ROI

A solo trainer pays back the platform in **week 2**.

26. Competitive Differentiation

Capability	Apollo	Lemlist	HubSpot	Buffer	Tagmango	Generic GPT	CorporateMind AI
HR discovery (intent-matched)	Partial	—	—	—	—	—	✓
Personalized outreach (per-recipient)	Partial	Partial	—	—	—	Partial	✓
Multi-agent orchestration	—	—	—	—	—	—	✓
Trainer-niche aware	—	—	—	—	Partial	—	✓
AI proposal generation	—	—	—	—	—	—	✓
Omnichannel (WA+TG+IG+FB+Email+LI-post)	Partial	—	Partial	Partial	Partial	—	✓
WhatsApp Business native	—	—	Partial	—	Partial	—	✓
Compliance-as-agent	—	—	—	—	—	—	✓
Self-healing workflows	—	—	—	—	—	—	✓
Indian pricing (₹999 entry)	—	—	—	—	✓	—	✓
Positioning	US-centric DB	Cold-email tool	Generic CRM	Social scheduler	Course platform	Chatbot	Autonomous B2B Growth OS

27. Pillar-Based Platform Architecture (7 Pillars)

Each Pillar follows the same 5-stage autonomous loop (\$9.2). The pattern is the moat: any future capability plugs in as a new Pillar using the same orchestrator + LLMOps spine.

PILLAR 1 — TRAINER INTELLIGENCE

What It Solves: Trainer profiles in current SaaS are 30-field forms abandoned mid-completion. Trainers know their craft; they cannot articulate it as metadata. The result: bad segmentation, generic outreach, low fit.

Workflow:

Upload (poster/video/PDF/voice) → OCR + Vision + Whisper → Field extraction
→ Embedding → Profile lock → Match-ready vector

Agents: TrainerProfileAgent, ingestion workers. **ROI:** Onboarding time 60+ min form → < 5 min upload. Activation rate +200%.

PILLAR 2 — HR DISCOVERY

What It Solves: LinkedIn manual scrolling, stale CSVs, irrelevant lists. Trainers waste 12–20 hrs/week on research.

Workflow:

Trainer profile vector → Match candidate companies (industry/size/region rules + semantic)
→ Surface HR/L&D contacts from opt-in sources → Score (0–1) per (trainer × contact)
→ Trainer reviews segment → Lock for campaign

Agents: HRDiscoveryAgent, ScraperWorker (ToS-bounded), SentimentScorer. **ROI:** 50+ matched contacts in < 30 min, 100% opt-in-verified.

PILLAR 3 — OUTREACH AI

What It Solves: Cold outreach reply rates are < 3% because messages are channel-blind, recipient-blind, and tone-blind.

Workflow:

Recipient × Trainer → Personalization context (industry, recent news, fit angle)
→ Channel-aware draft (Email/WA/Telegram/LI-post) → A/B variants
→ ComplianceGuard → HITL if > 200 → Send → Track delivery/open/reply

Agents: OutreachAgent (lead), CopyWriter, Scheduler, AttributionAgent. **ROI:** Reply rate 2% → 12–18%. Campaign cycle 3 days → 60 min.

PILLAR 4 — SOCIAL AUTOMATION

What It Solves: Trainers post random content randomly. No cadence, no brand consistency, no funnel.

Workflow:

Content calendar (auto-generated from trainer profile + campaign goals)
→ Channel-specific drafts (IG reel caption, FB post, LI post, TG broadcast)
→ Trainer approves → Scheduled publish → Engagement tracking → Optimizer learns

Agents: InstagramAgent, FacebookAgent, TelegramAgent, LinkedInAgent (publish only). **ROI:** 4× content output at same trainer time; consistent brand voice.

PILLAR 5 — PROPOSAL AUTOMATION

What It Solves: Proposals take 3–5 hours. Slow proposals lose deals.

Workflow:

Positive reply detected → Trainer triggers proposal → ProposalAgent drafts:
cover / problem / solution / agenda / pricing / case studies
→ PDF render → Trainer edits → Send → Tracked

Agents: ProposalAgent, PDFRenderer. **ROI:** Turnaround 4h → 5 min. Close rate +30%.

PILLAR 6 — CRM INTELLIGENCE

What It Solves: Generic CRMs require manual updating; trainers don't.

Workflow:

Every send / reply / open / meeting → auto-update lead state
→ Pipeline board reflects reality → Daily insight cards: who to nudge today

Agents: AnalyticsAgent, NextBestAction. **ROI:** Zero manual CRM upkeep; pipeline always current.

PILLAR 7 — MULTI-AGENT RUNTIME (LangGraph + Euri)

What It Solves: Single-model chatbots cannot orchestrate the above six. They lack memory, tools, gating, and observability.

Provides: Shared WorkflowState , checkpointing, HITL gates, tool sandboxing, ComplianceGuard, model routing, eval pipelines, replay debugging.

ROI: Enables Pillars 1–6 to exist as autonomous systems instead of GPT wrappers.

28. Self-Healing Workflow Systems

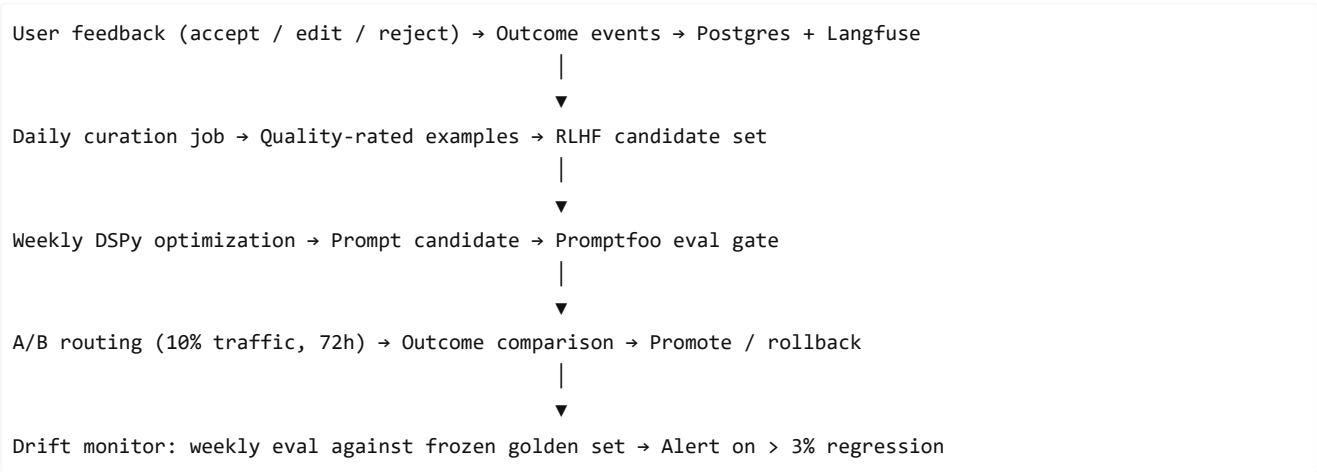
Failure is a first-class operating condition. Self-healing has six layers:

- 1. **Per-step retry with exponential backoff** for transient errors (network, 5xx, 429).
- 2. **Self-repair re-prompting** for schema-invalid LLM outputs (max 2 attempts, then escalate).
- 3. **Provider fallback chain** (Euri primary → secondary → tertiary → Ollama local → cached / HITL).
- 4. **Sandboxed tool execution** with per-tool timeouts, output validators, idempotency keys (safe replay).
- 5. **Checkpointed workflow state** so any node resumes after a process restart or HITL pause.
- 6. **DLQ + reaper + replay** to capture, group, diagnose, and replay failures once a fix ships.

Failure isolation: Every Celery worker pool bulkheaded by queue; one tenant's runaway loop cannot starve another's outreach send. Per-tenant concurrency caps enforce this at queue level.

Circuit breakers on every external dep: open after 5 consecutive failures or > 50% error rate in 60s; half-open after 60s; closed on 3 consecutive successes. Open-circuit responses are degraded but never silent — UI surfaces "AI temporarily unavailable; falling back to draft mode."

29. Continuous Learning Pipelines



The RLHF dataset is the **proprietary moat** — it compounds with every interaction, and cannot be replicated by a competitor without operating tenants of the same density and tenure.

30. Future Roadmap

Phase	Timeline	Deliverables	Success Metric
Phase 1: Cockpit	Month 0–3	Trainer Intel + HR Discovery + Outreach + Email + WA, single-workspace tenants	80 paying tenants
Phase 2: Omnichannel	Month 4–6	Telegram, IG, FB, LinkedIn-post, Proposal generation, multi-workspace	400 tenants, ₹15 L MRR

Phase 3: Optimization & White-Label	Month 7–9	CampaignOptimizer, semantic cache, prompt-opt loop, agency white-label v1	1,000 tenants, ₹45 L MRR
Phase 4: Enterprise & Network	Month 10–14	SSO, dedicated cluster, audit, cross-tenant insight network (opt-in)	1,600 tenants, ₹1.4 Cr ARR
Phase 5: Platform	Month 15–24	Voice outreach (inbound + outbound), AI meeting summaries, public tool/plugin SDK, marketplace	2,500+ tenants, ₹3 Cr ARR

Sprint-Level Milestones (Phase 1)

- **Sprint 1 (W1–2):** Monorepo, tenant + workspace + auth + onboarding wizard
- **Sprint 2 (W3–4):** Trainer upload + ingestion + TrainerProfileAgent + profile UI
- **Sprint 3 (W5–6):** HR Discovery agent + lead list UI + opt-in source registry
- **Sprint 4 (W7–8):** Outreach generation + Email channel + ComplianceGuard v1
- **Sprint 5 (W9–10):** WhatsApp channel + follow-up cadence + HITL approval queue
- **Sprint 6 (W11–12):** CRM/pipeline UI + insight cards + observability v1 + design-partner onboarding (10 tenants)

31. Risks & Assumptions

Risk	Likelihood	Impact	Mitigation
Frontier LLM API breakage / pricing shock	Medium	High	Euri abstraction + Ollama fallback + monthly cost ceilings
WhatsApp policy change blocks marketing templates	Medium	High	Multi-channel (Email + IG + Telegram) so WA is not single point of failure
LinkedIn ToS tightening on automation	High	Medium	Strict policy: only public company data + page posts; no personal DM automation ever
Indian email deliverability (DKIM/SPF/DMARC) issues	Medium	Medium	Per-tenant subdomain warmup; Postmark/SES with reputation monitoring
Trainer/coach segment too small for ₹2 Cr ARR	Low	High	Expand to consultants, sales enablement firms, leadership coaches (TAM × 4)
Competitor (Apollo, Lemlist, HubSpot) adds agent layer	Medium	Medium	Speed + Indian channel-native (WA/UPI) + niche specialization is our moat
Founder bandwidth as solo-dev MVP	High	High	Phase 1 scope intentionally bounded; outsource design/QA
Compliance incident (spam complaint, opt-in dispute)	Medium	High	ComplianceGuard as first-class agent; audit log on every send; DPDP-compliant erasure

Assumptions:

- Euri AI maintains ≥ 99.5% gateway availability.
- Frontier model token prices decline ~30% YoY.
- WhatsApp Business Cloud API remains the dominant Indian B2B messaging channel.
- Trainers accept HITL-gated automation (not full autonomy on Day 1).
- LinkedIn maintains public company-page API access on current terms.

32. Open Questions

1. Self-host Whisper for transcription at MVP, or use Deepgram free tier to ship 4 weeks earlier?

- 2. Should the cross-tenant "insight network" (Horizon 2) be opt-in or default with anonymization?
- 3. White-label: allow tenants to bring their own LLM keys (BYOK) and bypass our gateway?
- 4. Email provider for outbound: Postmark (deliverability), SES (cost), or Resend (DX)?
- 5. WhatsApp BSP: direct Cloud API or via AiSensy/Gupshup (cheaper templates, slower approvals)?
- 6. Where do we draw the line between "AI proposes" and "AI sends without approval" — per-feature default or tenant policy?
- 7. Phone-OTP via MSG91 (Indian) or Twilio Verify (global)?
- 8. Multi-region: second region after India — SEA (volume) or Middle East (avg deal size)?
- 9. Do we publish RLHF-derived insights back to tenants ("trainers like you saw X work") as a feature?
- 10. Voice outreach (Phase 5) — partner with Vapi/Bland, or extend in-house from RevenueTable AI's stack?
- 11. Pricing: per-workspace flat vs. per-send usage vs. hybrid?
- 12. SOC2: pre-revenue (Enterprise readiness) or post-Series-A (capital efficient)?
- 13. Do we offer a "managed migration" service from Apollo/Lemlist as a paid wedge?
- 14. Native mobile app vs. PWA + WhatsApp as distribution?
- 15. How is "trainer/coach" distinguished from "consultant/freelancer" in pricing — same product, different acquisition channel?
- 16. Public stance on AI-generated reviews/testimonials and inbound UGC moderation?
- 17. Long-term: monetize anonymized hiring-intent signals as a B2B data product to L&D vendors?

33. Enterprise Expansion Strategy

- **Agency Rollout Playbook:** Pilot with 1 lead trainer at the agency → 60-day metrics review → batch-onboard remaining trainers with shared HR DB + brand guardrails; HQ approves brand-level templates; per-trainer autonomy within those guardrails.
- **White-Label:** Re-brandable theme, custom domain, BYOK (Enterprise+), sub-tenant management UI for the reseller, revenue share or wholesale pricing.
- **LMS Partner Integrations:** Tagmango, Exly, Graphy — bidirectional sync of trainer profiles + booked workshops.
- **Marketplace (Horizon 3):** Third-party agents (e.g., "Diwali campaign pack", "Q1 L&D budget cycle pack", "GST proposal pack") sold via marketplace; revenue share with publishers.
- **Data Residency & Sovereignty:** Region-pinned tenants; India region default; expansion to SEA (Singapore) and ME (Bahrain) by Phase 5.

34. Final Strategic Positioning

CorporateMind AI is **not** competing for "best email tool" or "best CRM" slot. Those categories are commoditized races to zero margin. We are creating — and intend to own — the category of **Autonomous AI Business Growth Operating System for Expert-Services Professionals**, where the trainer's role shifts from orchestrator to approver-and-deliverer, and where every Pillar plugged into the LangGraph spine compounds the moat:

more agents → more outcomes → better RLHF → cheaper inference → higher margin → faster shipping → more agents

The technical bet: a tightly-scoped 12-agent system, on a free-OSS-first stack, with provider-abstracted inference and ruthless cost telemetry, can deliver investor-grade unit economics from tenant #1 while remaining a single-engineer-feasible MVP.

The product bet: trainers and coaches — once they have one workshop booked through CorporateMind in week 2 — will never voluntarily return to Apollo + Buffer + spreadsheets.

Every architectural choice in this document is in service of that two-part bet: **ship fast on a defensible substrate, evolve the substrate without rewriting it.**

Appendix A — AI Memory Architecture (Deep Spec)

A.1 Memory Class Definitions

Class	Definition	Backing	Retrieval Pattern
Working	Per-workflow-run scratchpad	In-process Pydantic state	Direct attribute access

Short-term Conversational	Last N turns of a chat session	Redis hash, conv:{session_id}, TTL 30m sliding	Full-load on turn
Episodic	"What happened" — past agent runs and outcomes	Postgres agent_runs, agent_events; vectors in Qdrant campaign_outcomes_{org}	Vector search + recency-weighted rerank
Semantic	"What is true" — trainer expertise, HR/company facts	Qdrant per-tenant; primary in Postgres	Hybrid search + cross-encoder rerank
Procedural	"How to do X" — versioned prompts and skills	prompt_templates table	Template fetch by (name, version, env)

A.2 Retrieval Scoring

`final_score =  $\alpha$  · cosine +  $\beta$  · BM25 +  $\gamma$  · recency_decay +  $\delta$  · outcome_weight` Default `( $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ) = (0.55, 0.20, 0.15, 0.10)` — tunable per agent.

A.3 Pruning & TTL

- Conversational: hard 30-min sliding TTL.
- Episodic: 90 days hot in Postgres, archived to S3 Parquet cold; vectors retained 12 months.
- Semantic: re-embed quarterly; old vectors archived; never destructively deleted within retention window.
- Procedural: versioned, never deleted (audit trail).

A.4 Memory Observability

Every retrieval emits a Langfuse span with: query, top-k IDs, scores, latency, tokens-injected. A "Memory Inspector" admin view shows what context any past run saw.

A.5 Cost Discipline

- Memory-injected tokens budgeted per run (default 2,400 tokens).
- Truncation order: oldest episodic > lowest-scored semantic > lowest-scored conversational.
- Cache-aware: identical retrieval queries within a run hit Redis-side memo.

Appendix B — AI Agent Runtime System (Deep Spec)

B.1 Lifecycle

```
INIT → HYDRATE_CONTEXT → PLAN → (EXECUTE ↔ VERIFY)* → FINALIZE → EMIT_EVENTS → ARCHIVE
                                     |
                                     └ on fail → RETRY / FALLBACK / HITL / DLQ
```

B.2 Planner-Executor Split

- **Planner** generates a strict-schema `Plan { steps: List[Step] }` with explicit `tool`, `inputs`, `expected_output_schema`.
- **Executor** is a deterministic dispatcher — it does not reason; it executes the plan. This split makes every action auditable and replayable.

B.3 Checkpointing

Every state transition writes `(workflow_id, node, version, state_blob, ts)` to `workflow_checkpoints`. Workflows are resumable from any checkpoint, in any process. Resumption validates `schema_version`; migrations registered per `(from, to)`.

B.4 Task Delegation

The RootOrchestrator decomposes high-level intent into sub-agent invocations. Sub-agents are first-class: own state, memory, tools. They communicate **only** via the shared WorkflowState , never directly — which makes the topology a tree (deterministic, debuggable), not a mesh.

B.5 Concurrency

- Per-tenant concurrency cap on agents Celery queue (default: Starter=2, Growth=8, Enterprise=32).
- Within a workflow, parallel sub-agent calls fan out via asyncio.gather with per-call timeouts.

B.6 Tool Sandboxing

Tools registered with explicit schemas + per-tool budgets (max_calls_per_run , max_tokens_per_call , timeout_s). Executor enforces. Tools marked read_only (default), mutating_low_risk , or mutating_high_risk — the last requires HITL or explicit policy approval.

B.7 Runtime Observability

Each node emits an OTEL span carrying tenant_id , agent , node , model , tokens_in , tokens_out , latency_ms , result_status , request_id . Langfuse correlates the LLM trace; Sentry captures any exception with the same request_id .

Appendix C — Event Catalog & Contracts

C.1 Event Envelope

```
{
  "event_id": "01J4...",
  "event_type": "outreach.message_sent",
  "version": "1.0",
  "occurred_at": "2026-05-23T14:32:11.482Z",
  "tenant_id": "org...",
  "workspace_id": "ws...",
  "actor": { "type": "agent", "id": "OutreachAgent" },
  "payload": { "...": "..." },
  "trace_id": "..."
}
```

C.2 Naming Convention

<domain>.<entity>.<verb_past> — e.g. lead.discovered , outreach.message_sent , campaign.approval_requested .

C.3 Core Events

Event	Producer	Consumers
trainer.profile_extracted	TrainerProfileAgent	HR discovery trigger, dashboard
lead.discovered	HRDiscoveryAgent	Enrichment, dashboard
outreach.draft_ready	OutreachAgent	HITL approval notifier
outreach.message_sent	Send pipeline	Attribution, analytics
outreach.replied	Reply ingester	Conversation thread, NextBestAction
campaign.approval_requested	OutreachAgent	Trainer inbox
proposal.draft_ready	ProposalAgent	Trainer review

meeting.scheduled	Calendly webhook	CRM, reminder scheduler
agent.run_failed	LLMOpsGuardian	DLQ persister, SRE alert
compliance.block	ComplianceGuardAgent	Audit log, trainer notification

C.4 Delivery Semantics

At-least-once, idempotent on `event_id`. Consumers register handlers with explicit `is_idempotent: true` declaration — non-idempotent handlers rejected at registration.

C.5 Replay

Event log append-only; per-consumer cursor in Redis. Replay is consumer-scoped: rewind cursor to a timestamp; consumer re-processes. Replays gated by feature flag per environment.

Appendix D — Multi-Tenancy Reference Model

```
Organization (LeadIn Coaching LLP – agency)
├─ Workspace: Priya NLP (lead trainer)
│   ├── Trainer profile: Priya
│   └── Trainers: Priya, Asha (junior)
├─ Workspace: Rajiv Leadership (sister brand)
│   └── Trainer: Rajiv
└─ Settings
    ├── Subscription: Growth × 2 workspaces
    ├── AI Budget: ₹4,000/mo
    ├── Feature Flags: linkedin_post=on, proposal_v2=on, semantic_cache=on
    └── Compliance: DPDP region=India, retention=24mo
```

- Per-workspace brand voice, brand assets, template library.
- Per-workspace channels (each trainer connects own WA/IG/FB).
- Per-org billing, RBAC, AI budget, feature flags, data residency.
- Per-user role (scope = org / workspace / specific resource).

This hierarchy is the **single source of truth** referenced by every service, agent, and policy. Every other capability — from semantic memory partitioning to inference rate-limits — derives from it.

Appendix E — Proposed `CLAUDE.MD` (engineering operating manual)

To be written to `./CLAUDE.MD` in Phase 0. Adapted from the AI Customer Support Copilot reference for CorporateMind AI's actual scope.

```
# CLAUDE.md – CorporateMind AI

## Role
Act as a staff engineer, AI architect, and SaaS production reviewer for CorporateMind AI – an autonomous AI corporate-outreach and multi-channel growth OS for trainers, coaches, consultants, and speakers.

## Project context
- Frontend: Next.js 14 (App Router, RSC, TanStack Query, Tailwind, shadcn/ui)
- Backend: FastAPI (async) – modular monolith
- Data: PostgreSQL (primary), Redis (cache/queue/rate-limit), Qdrant (embeddings/semantic cache)
- AI runtime: LangGraph multi-agent
- AI gateway: Euri AI Gateway – the ONLY egress to LLM providers
```

- Workers: Celery (Redis broker) + Celery beat
- Storage: Cloudinary / S3-compatible
- Channels: Email, WhatsApp Business Cloud API, Telegram, Instagram Graph, Facebook Graph, LinkedIn (public-only)
- Deploy: Vercel (web) + Railway (api/workers/db/redis)
- Observability: Langfuse, Prometheus + Grafana, Sentry, OpenTelemetry

Architecture (modular monolith, 7 pillars)

Modules under `apps/api/src/corpmind/modules/`:

`identity, trainer\_intel, hr\_discovery, outreach, social, whatsapp, proposals, crm, campaigns, analytics, billing, compliance`.

Each module follows Ports & Adapters: `api.py | service.py | repo.py | models.py | schemas.py | events.py`. Inter-module rule: modules MUST NOT import each other's `repo.py` or `models.py`. Cross-module talk via service interfaces (DI) or in-process event bus.

Core behavior

- Production-grade code over quick hacks. No demo scaffolding.
- Preserve patterns; don't refactor opportunistically.
- Inspect neighbors before editing.
- Briefly justify tradeoffs when introducing new patterns.
- Design for horizontal scale and per-tenant isolation from day one.
- Stage 1 = modular monolith on Railway. Do NOT preemptively introduce Kafka, k8s, or microservices.

Multi-tenancy rules (NON-NEGOTIABLE)

- Every business table has `tenant\_id UUID NOT NULL` + composite index.
- `TenantContext` contextvar set in middleware; every query MUST filter by it.
- Postgres RLS enabled as defense-in-depth.
- Qdrant filters include `tenant\_id` predicate on every search.
- Cross-tenant access is a P0 bug; regression test per new table.

Backend engineering rules

- Async FastAPI endpoints unless strong reason not to.
- Route handlers ≤ 15 lines; logic in `service.py`, persistence in `repo.py`.
- Pydantic v2 for ALL schemas – no raw dicts cross boundaries.
- SQLAlchemy 2.0 async + Alembic. Never edit applied migrations.
- Structured JSON logging (structlog); correlation IDs propagated.
- Explicit exception handling; stable error envelope.
- No hardcoded secrets/URLs – `pydantic-settings` only.
- DI via FastAPI `Depends()`; no global state.
- Idempotency-Key on all mutating endpoints (Redis, 24h TTL).

Frontend engineering rules

- Next.js App Router, RSC by default.
- Strict TypeScript – no `any`.
- Small reusable components; feature-sliced layout.
- Separate UI from data fetching (TanStack Query hooks).
- Every async surface has loading, empty, error states.
- API client centralized; types generated from OpenAPI.
- SSE hook for agent-run / campaign-progress streams.

Database rules

- UUID v7 primary keys.
- Indexes on hot paths.
- JSONB + GIN for flexible metadata; never for fields queried by equality often.
- Soft-delete for customer-facing entities.

- Partition `outreach\_messages`, `whatsapp\_messages`, `analytics\_daily` monthly at scale.
- No destructive schema changes without reversible migration.

Euri AI Gateway rules (sole LLM egress)

- All LLM calls via `corpmind.ai.euri\_client.EuriClient`. Importing `openai`, `anthropic`, `google.generativeai` directly is forbidden.
- `routing.py` maps `task\_type` → (primary, fallback_chain).
- Wrap every call: `PromptInjectionFilter` → `PIIRedactor` → `SemanticCache` → `Euri` → `OutputModerator`.
- Every call traced to Langfuse with `tenant\_id`, `agent`, `prompt\_version`, `cost\_tokens`.
- Prompts versioned (file-backed registry); never inline a prompt in business logic.
- Token budgets enforced per tenant; over-budget returns typed error.

AI agent rules (LangGraph)

- One responsibility per agent. Tools explicit and minimal.
- Shared `AgentState` TypedDict; no untyped graph state.
- Deterministic helpers around fragile model behavior.
- Fallback path for low-confidence outputs.
- Never expose raw model output without post-processing + moderation.
- HITL interrupts when: recipients > 200, enterprise tier, ComplianceGuard flag, sensitive content classifier.
- Log key decisions to Langfuse.

Semantic cache rules

- Qdrant `prompt\_cache` keyed by (tenant_id, task_type, embedding).
- Cache only deterministic-by-input tasks. Never cache personalized outreach copy.
- TTL + invalidation on prompt-version bump.

RAG / retrieval rules

- Separate ingestion → chunking → embedding → retrieval → generation.
- Chunk size/overlap configurable per source.
- Source attribution on every retrieval result.
- Log low-confidence retrievals.

Campaign & lead lifecycle rules

- Campaign states: `draft` → `pending\_approval` → `scheduled` → `sending` → `completed` | `paused` | `failed`.
- Lead pipeline: `discovered` → `contacted` → `engaged` → `replied` → `meeting\_scheduled` → `meeting\_done` → `booked` | `lost`.
- Auto-classification user-overridable.
- Duplicate-detection logs matches + confidence.
- Routing rules configurable in DB, not code.

Channel adapter rules

- Every channel implements `corpmind.channels.base.ChannelAdapter` (`send`, `fetch\_status`, `handle\_webhook`).
- No provider SDKs in business modules.
- Circuit breakers + exponential backoff.
- All outbound calls logged with status + latency.
- Webhooks verify HMAC BEFORE parsing.
- Per-channel Redis token-bucket rate limiters tuned to provider limits.

Compliance & anti-spam rules (NON-NEGOTIABLE)

- Every outbound message via `ComplianceGuardAgent` before send.
- Opt-ins tracked per (tenant, contact, channel); unsubscribes global per tenant.
- WhatsApp: enforce 24-hour window; approved templates outside.
- Email: physical address + unsubscribe footer; honor `List-Unsubscribe`.
- LinkedIn: NEVER automate personal DMs or scrape private data. Public company data + opt-in only.

- Duplicate detection (content + recipient) before send.
- Frequency cap per (tenant, contact, channel) – default 1/week.
- Audit log every send + every block.

Integration rules

- Third-party via adapter pattern; no direct SDK calls in business modules.
- Endpoints config-driven.
- Circuit breakers + retries with jitter.
- Secrets in Doppler/Infisical for prod.

Security rules

- Validate + sanitize ALL untrusted input.
- Never expose stack traces.
- JWT (RS256), refresh rotation; httpOnly cookies.
- RBAC via Casbin.
- PII detection + masking in logs.
- Treat uploaded posters/videos and scraped HR data as untrusted; sandbox parsing.
- Encrypt sensitive columns at rest (channel tokens, OAuth refresh tokens).
- Audit log for every privileged operation; append-only.
- GDPR + DPDP-compliant; per-tenant data export + deletion.
- Prompt injection: scrub retrieved/user content before LLM call.

Architecture Decision Records (ADR)

- Significant change → ADR in `docs/adr/NNNN-title.md`.
- ADRs immutable after acceptance; change by writing superseding ADR.
- Significant: DB strategy, AI gateway, queue, tenancy, observability, deploy topology, agent topology, channel-adapter contracts, billing model.
- ADR states: context, decision, alternatives, consequences, date.

Feature flags

- All risky/user-visible features ship behind a flag.
- AI behavior changes support gradual rollout (% of tenants).
- New agent workflows ship with kill-switch.
- Experimental prompts/models tenant-gated before general rollout.
- Flags have owner + expiry; stale flags removed quarterly.

AI cost governance

- Every agent run records: `prompt_tokens`, `completion_tokens`, `latency_ms`, `estimated_cost_inr`, `model`, `cached`. Mirrored to Langfuse.
- Hard budget ceiling per tenant per period; exceed → `BudgetExceededError``.
- Soft thresholds 70/85/95% → events + UI banner.
- Expensive workflows (>500-recipient outreach, full proposal gen) require explicit UI confirmation.
- Small models for: classification, extraction, ranking. Premium for: proposals, personalized outreach, viral hooks.
- Cache hit rate is a tracked SLO.

Queue & worker safety (Celery)

- Every task idempotent (``task_key`` deterministic).
- Declares: ``max_retries``, ``retry_backoff``, ``time_limit``, ``soft_time_limit``, ``acks_late=True``.
- DLQ for exhausted retries; alert on depth.
- Emits Prometheus metrics: ``task_duration_seconds``, ``task_outcome{status}``, ``task_retries_total``.
- Long workflows checkpoint to Postgres.
- Fan-out chunks recipients; self-throttle to channel limits.
- Queue-depth circuit breaker: defer new campaigns above threshold.
- Separate queues: ``ingestion``, ``outreach``, ``social``, ``analytics``.

Prompt engineering standards

- Prompts in `apps/api/src/corpmind/ai/prompts/` (versioned files); loaded via registry – never inlined.
- Each prompt declares: role, constraints, input schema, output schema (JSON), safety rules, examples.
- Prefer structured output (JSON mode / function calling); avoid hidden chain-of-thought in prod.
- Each prompt has: semver, regression suite (fixtures + Langfuse evals), tracked metrics, rollback.
- Promotion via feature-flag flow: shadow → % rollout → 100%.
- Prompt diffs go through code review.

Incident response

- SEV ladder: SEV1 (data loss / cross-tenant leak / outage), SEV2 (major feature down, mass-send failure, billing broken), SEV3 (degraded UX), SEV4 (cosmetic).
- SEV1/SEV2 → page on-call, incident channel, IC assigned, status-page update within 15 min.
- Updates every 30 min (SEV1) / hourly (SEV2).
- Stop the bleeding first: kill-switch, pause queues, rotate keys – before RCA.
- Runbooks in `ops/runbooks/` per failure mode (Euri down, WA template storm, queue backlog, DB connection exhaustion, mass-send compliance flag).
- Blameless postmortem within 5 business days; architectural root cause → ADR.
- Blast-radius check in first 10 min: tenant-isolated or platform-wide?
- Customer-data incidents trigger DPDP/GDPR breach-notification workflow (legal owner, 72h clock).

Deployment guardrails

- Trunk-based; `main` always deployable; squash-merge.
- CI gates: ruff, mypy, eslint, tsc, pytest, alembic upgrade check, OpenAPI diff, container build.
- Migrations deploy BEFORE app code that depends; every migration reversible; destructive changes two-step.
- Environments: dev → preview → staging → prod. No direct prod deploys; staging soak ≥ 1h for risky changes.
- Progressive rollout: blue/green or canary; new agent graphs / channel adapters 5% → 25% → 100% behind flag.
- Auto-rollback on SLO burn (error rate, p95, LLM fallback) within first 15 min.
- No prod deploys Fridays after 14:00 IST, weekends, or during known customer campaigns.
- Secrets never logged; secret-scanning per PR; rotation scheduled.
- Long migrations gated by maintenance flag, run via one-off worker with `lock\_timeout` and `statement\_timeout`.
- Every new integration ships with kill-switch flag + rollback plan.
- Pre-deploy PR template: migrations? flag? rollback? Langfuse dashboard? alerts reviewed?

Testing rules

- Unit tests per `service.py` function.
- API tests (httpx + testcontainers) per endpoint.
- Integration tests for multi-step flows (upload → extract → discover → generate).
- AI: fixture-based assertions on shape/schema, NOT exact text match.
- Tenant-isolation regression test per new table.
- Don't mark complete without running ruff, mypy, pytest, eslint, tsc.

Observability rules

- Structured JSON logs; request-id + tenant-id + run-id in every line.
- OTel traces span FastAPI → Celery → Euri → Qdrant.
- Langfuse for LLM traces: prompt version, model, tokens, latency, cost, eval scores.
- Prometheus: HTTP RED, queue depth, channel send rate, token spend per tenant.
- Alerts: SLA breach, error rate > 1%, fallback rate > 5%, queue backlog > 1k.

Review checklist (before merge)

- Module boundaries respected.
- `tenant\_id` propagated and filtered on every query.
- LLM calls via `EuriClient`; prompt versioned.
- Outbound messages pass through `ComplianceGuardAgent`.

- Idempotency keys handled on mutating endpoints.
- Edge cases: empty, timeouts, rate limits, partial failures.
- Structured logs + Langfuse trace for important flows.
- Tests added/updated; tenant-isolation test if new table.
- No PII in logs.
- No direct provider SDKs imported.
- Migration reversible; no destructive change without backup.
- Safe to deploy: feature-flagged if user-visible.

Appendix F — Step-by-Step Implementation Roadmap

Phase 0 — Foundations (Week 1)

Monorepo (`apps/api` , `apps/web` , `packages/` , `infra/`); ruff + mypy + pre-commit; ESLint + Prettier; dev compose (PG/Redis/Qdrant); CI (lint + test + migrate-check); empty FastAPI + Next.js deployed to Railway/Vercel; write `CLAUDE.MD` (from Appendix E); ADR-0001 (modular monolith), ADR-0002 (Euri as sole LLM egress).

Phase 1 — Identity & Multi-Tenancy (Week 2)

`orgs` , `workspaces` , `users` , `memberships` , `api_keys` ; signup/login/refresh; JWT; Casbin RBAC; TenantContext middleware; RLS policies; frontend auth + protected dashboard shell.

Phase 2 — AI Plane (Week 3)

`EuriClient` with routing matrix + fallback; prompt registry (file-backed); semantic cache (Qdrant `prompt_cache`); guardrails (injection filter, PII redactor, output moderation); Langfuse integration; LangGraph runtime + `AgentState` ; RootOrchestrator skeleton.

Phase 3 — Trainer Intelligence (Week 4)

Cloudinary upload API; ingestion Celery tasks (OCR, transcription); `TrainerProfileAgent` extracts profile; Qdrant `trainer_profiles_{org}` ; upload + profile-review UI.

Phase 4 — HR Discovery (Week 5)

`companies` , `hr_contacts` ; `HRDiscoveryAgent` (web.search + `companies.lookup` , opt-in only); semantic match trainer ↔ company; lead-list builder UI; CSV export; dedupe.

Phase 5 — Outreach + Email Channel (Week 6)

`OutreachAgent` generates A/B variants; Email adapter (Postmark/SES/Resend); campaigns + recipients + messages tables; ComplianceGuard v1; campaign builder UI + approval-gate flow.

Phase 6 — WhatsApp Channel (Week 7)

WA Business Cloud adapter (templates, sessions); opt-in tracking, 24h window; webhooks; `WhatsAppAgent` follow-ups; WA campaign UI; template management.

Phase 7 — CRM, Meetings, Proposals (Week 8)

Conversation threading; pipeline Kanban; Calendly/Google Meet webhooks → meetings; `ProposalAgent` generates pitch deck (markdown → PDF via WeasyPrint).

Phase 8 — Telegram + Instagram + Facebook + LinkedIn-post (Weeks 9–10)

One channel per sprint week via same `ChannelAdapter` ; scheduling via Celery beat; per-channel tone tuning.

Phase 9 — Analytics, Optimizer, Compliance v2 (Week 11)

`analytics_daily` rollups; Grafana + in-app dashboards; CampaignOptimizer nightly job; ComplianceGuard v2 (dup detection, frequency cap, classifier); HITL approval queue UI.

Phase 10 — Billing, Limits, Polish (Week 12)

Razorpay subscriptions (₹999/₹4,999/Custom); usage counters; per-tenant budgets; white-label theming hooks; k6 load test to 1K concurrent tenants; SLO verification; security review (OWASP top-10).

Post-MVP (Phase 11+)

- WhatsApp AI chatbot (inbound replies by agent)
- Voice outreach (Vapi/Bland integration, reuses RevenueTable AI patterns)
- DSPy nightly prompt optimization at scale
- Extract WhatsApp & Social as separate services (Stage 2 trigger)

Critical Files (created in Phase 0)

- `apps/api/src/corpmind/main.py` — FastAPI app factory
- `apps/api/src/corpmind/core/config.py` — pydantic-settings
- `apps/api/src/corpmind/core/db.py` — async SQLAlchemy
- `apps/api/src/corpmind/ai/euri_client.py` — sole LLM egress
- `apps/api/alembic/env.py` — migration env
- `apps/web/app/layout.tsx` — Next.js root
- `infra/docker/compose.dev.yml` — local stack
- `.github/workflows/ci.yml` — lint/test/migrate
- `docs/adr/0001-modular-monolith.md`
- `docs/adr/0002-euri-as-sole-llm-egress.md`
- `CLAUDE.MD` (from Appendix E)

Verification Plan

End-to-end smoke after each phase:

1. `docker compose up` → PG/Redis/Qdrant up; `make dev` → api+worker+web up.
2. Trainer flow: upload poster → extracted profile in UI within 30s.
3. HR discovery: trainer profile → ≥ 10 matching opt-in HR leads.
4. Outreach: generate + approve → emails sent → tracked in CRM.
5. Compliance: > 200-recipient campaign blocks until manual approval.
6. Observability: Langfuse shows full agent trace; Grafana shows token spend.
7. Tenant isolation: two tenants, verify queries cannot cross (automated + manual).
8. Load: k6 to 100 RPS, p95 < 400ms, error < 0.5%.

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